

April 23, 2019

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Biosolids
Pace Project No.: 50221685

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on April 09, 2019. The results relate only to the samples included in this report. Results reported herein conform to the most current, applicable TNI/NELAC standards and the laboratory's Quality Assurance Manual, where applicable, unless otherwise noted in the body of the report.

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Brian Hall for
Melanie Booms
melanie.booms@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

CERTIFICATIONS

Project: Biosolids

Pace Project No.: 50221685

Indiana Certification IDs

7726 Moller Road, Indianapolis, IN 46268

Illinois Certification #: 200074

Indiana Certification #: C-49-06

Kansas/NELAP Certification #: E-10177

Kentucky UST Certification #: 80226

Kentucky WW Certification #: 98019

Michigan Department of Environmental Quality, Laboratory
#9050

Ohio VAP Certification #: CL0065

Oklahoma Certification #: 2018-101

Texas Certification #: T104704355

West Virginia Certification #: 330

Wisconsin Certification #: 999788130

USDA Soil Permit #: P330-16-00257

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

SAMPLE SUMMARY

Project: Biosolids

Pace Project No.: 50221685

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50221685001	Digester #4	Solid	04/08/19 13:05	04/09/19 09:48

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

SAMPLE ANALYTE COUNT

Project: Biosolids
Pace Project No.: 50221685

Lab ID	Sample ID	Method	Analysts	Analytes Reported
50221685001	Digester #4	EPA 8081	KAV	21
		EPA 8081	KAV	8
		EPA 8082	RID	8
		EPA 8151A	BJW	3
		EPA 6010	RAM	6
		EPA 6010	JPk	8
		EPA 6020	DMT	10
		EPA 7470	LBT	1
		EPA 7471	LBT	1
		EPA 8270	JCM	67
		EPA 8270	JCM	18
		EPA 8260	RSW	55
		EPA 5030/8260	TMW	13
		SM 2540G	MMS	1
		SM 2540G	MMS	1
		SM 2540G	MMS	1
		EPA 9038	TPD	1
		EPA 9045	SCM	1
		EPA 351.2	ZM	1
		EPA 353.2	SLB	2
		SM 4500-NH3 G	GWA	1
		SM 4500-Cl-E	GWA	1

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50221685

Sample: Digester #4 **Lab ID: 50221685001** Collected: 04/08/19 13:05 Received: 04/09/19 09:48 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8081 GCS Pesticide Solids Analytical Method: EPA 8081 Preparation Method: EPA 3546								
Aldrin	<433	ug/kg	433	1	04/12/19 11:35	04/15/19 16:30	309-00-2	P1
alpha-BHC	<433	ug/kg	433	1	04/12/19 11:35	04/15/19 16:30	319-84-6	
beta-BHC	<433	ug/kg	433	1	04/12/19 11:35	04/15/19 16:30	319-85-7	
delta-BHC	<433	ug/kg	433	1	04/12/19 11:35	04/15/19 16:30	319-86-8	
gamma-BHC (Lindane)	<433	ug/kg	433	1	04/12/19 11:35	04/15/19 16:30	58-89-9	
Chlordane (Technical)	<8660	ug/kg	8660	1	04/12/19 11:35	04/15/19 16:30	57-74-9	
alpha-Chlordane	<433	ug/kg	433	1	04/12/19 11:35	04/15/19 16:30	5103-71-9	
gamma-Chlordane	<433	ug/kg	433	1	04/12/19 11:35	04/15/19 16:30	5103-74-2	
4,4'-DDD	<866	ug/kg	866	1	04/12/19 11:35	04/15/19 16:30	72-54-8	
4,4'-DDE	<866	ug/kg	866	1	04/12/19 11:35	04/15/19 16:30	72-55-9	
4,4'-DDT	<866	ug/kg	866	1	04/12/19 11:35	04/15/19 16:30	50-29-3	
Dieldrin	<866	ug/kg	866	1	04/12/19 11:35	04/15/19 16:30	60-57-1	
Endosulfan I	<433	ug/kg	433	1	04/12/19 11:35	04/15/19 16:30	959-98-8	
Endosulfan II	<866	ug/kg	866	1	04/12/19 11:35	04/15/19 16:30	33213-65-9	
Endosulfan sulfate	<866	ug/kg	866	1	04/12/19 11:35	04/15/19 16:30	1031-07-8	
Endrin	<866	ug/kg	866	1	04/12/19 11:35	04/15/19 16:30	72-20-8	
Endrin aldehyde	<866	ug/kg	866	1	04/12/19 11:35	04/15/19 16:30	7421-93-4	
Heptachlor	<433	ug/kg	433	1	04/12/19 11:35	04/15/19 16:30	76-44-8	
Heptachlor epoxide	<433	ug/kg	433	1	04/12/19 11:35	04/15/19 16:30	1024-57-3	
Toxaphene	<8660	ug/kg	8660	1	04/12/19 11:35	04/15/19 16:30	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	62	%.	18-136	1	04/12/19 11:35	04/15/19 16:30	2051-24-3	

8081 GCS Pest RV, TCLP

Analytical Method: EPA 8081 Preparation Method: EPA 3510

Leachate Method/Date: EPA 1311; 04/17/19 13:50 Initial pH: 7.91; Final pH: 5.09

gamma-BHC (Lindane)	<0.00025	mg/L	0.00025	1	04/22/19 08:58	04/22/19 15:45	58-89-9	1d
Chlordane (Technical)	<0.0025	mg/L	0.0025	1	04/22/19 08:58	04/22/19 15:45	57-74-9	1d
Endrin	<0.00050	mg/L	0.00050	1	04/22/19 08:58	04/22/19 15:45	72-20-8	1d
Heptachlor	<0.00025	mg/L	0.00025	1	04/22/19 08:58	04/22/19 15:45	76-44-8	1d
Heptachlor epoxide	<0.00025	mg/L	0.00025	1	04/22/19 08:58	04/22/19 15:45	1024-57-3	1d
Methoxychlor	<0.0025	mg/L	0.0025	1	04/22/19 08:58	04/22/19 15:45	72-43-5	1d
Toxaphene	<0.0050	mg/L	0.0050	1	04/22/19 08:58	04/22/19 15:45	8001-35-2	1d
Surrogates								
Decachlorobiphenyl (S)	37	%.	18-118	1	04/22/19 08:58	04/22/19 15:45	2051-24-3	

8082 GCS PCB Solids

Analytical Method: EPA 8082 Preparation Method: EPA 3546

PCB-1016 (Aroclor 1016)	<8430	ug/kg	8430	1	04/14/19 14:31	04/15/19 07:25	12674-11-2	
PCB-1221 (Aroclor 1221)	<8430	ug/kg	8430	1	04/14/19 14:31	04/15/19 07:25	11104-28-2	
PCB-1232 (Aroclor 1232)	<8430	ug/kg	8430	1	04/14/19 14:31	04/15/19 07:25	11141-16-5	
PCB-1242 (Aroclor 1242)	<8430	ug/kg	8430	1	04/14/19 14:31	04/15/19 07:25	53469-21-9	
PCB-1248 (Aroclor 1248)	<8430	ug/kg	8430	1	04/14/19 14:31	04/15/19 07:25	12672-29-6	
PCB-1254 (Aroclor 1254)	<8430	ug/kg	8430	1	04/14/19 14:31	04/15/19 07:25	11097-69-1	
PCB-1260 (Aroclor 1260)	<8430	ug/kg	8430	1	04/14/19 14:31	04/15/19 07:25	11096-82-5	
Surrogates								
Tetrachloro-m-xylene (S)	72	%.	26-140	1	04/14/19 14:31	04/15/19 07:25	877-09-8	

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50221685

Sample: Digester #4 **Lab ID: 50221685001** Collected: 04/08/19 13:05 Received: 04/09/19 09:48 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8151A CI Acid Herbicides TCLP								
Analytical Method: EPA 8151A Preparation Method: EPA 8151								
Leachate Method/Date: EPA 1311; 04/17/19 13:50 Initial pH: 7.91; Final pH: 5.09								
2,4-D	<0.0050	mg/L	0.0050	1	04/18/19 09:41	04/18/19 21:23	94-75-7	
2,4,5-TP (Silvex)	<0.0050	mg/L	0.0050	1	04/18/19 09:41	04/18/19 21:23	93-72-1	
Surrogates								
2,4-DCAA (S)	84	%	10-156	1	04/18/19 09:41	04/18/19 21:23	19719-28-9	
6010 MET ICP								
Analytical Method: EPA 6010 Preparation Method: EPA 3050								
Aluminum	4780	mg/kg	1920	1	04/16/19 16:52	04/18/19 13:30	7429-90-5	
Calcium	28900	mg/kg	1920	1	04/16/19 16:52	04/18/19 13:30	7440-70-2	
Magnesium	5070	mg/kg	1920	1	04/16/19 16:52	04/18/19 13:30	7439-95-4	
Phosphorus	39300	mg/kg	1920	1	04/16/19 16:52	04/18/19 13:30	7723-14-0	N2
Potassium	4570	mg/kg	1920	1	04/16/19 16:52	04/18/19 13:30	7440-09-7	
Sodium	6770	mg/kg	1920	1	04/16/19 16:52	04/18/19 13:30	7440-23-5	
6010 MET ICP, TCLP								
Analytical Method: EPA 6010 Preparation Method: EPA 3010								
Leachate Method/Date: EPA 1311; 04/17/19 13:50 Initial pH: 7.91; Final pH: 5.09								
Arsenic	<0.10	mg/L	0.10	1	04/18/19 13:45	04/18/19 23:07	7440-38-2	
Barium	<5.0	mg/L	5.0	1	04/18/19 13:45	04/18/19 23:07	7440-39-3	
Cadmium	<0.050	mg/L	0.050	1	04/18/19 13:45	04/18/19 23:07	7440-43-9	
Chromium	<0.10	mg/L	0.10	1	04/18/19 13:45	04/18/19 23:07	7440-47-3	
Lead	<0.10	mg/L	0.10	1	04/18/19 13:45	04/18/19 23:07	7439-92-1	
Selenium	<0.10	mg/L	0.10	1	04/18/19 13:45	04/18/19 23:07	7782-49-2	
Silver	<0.10	mg/L	0.10	1	04/18/19 13:45	04/18/19 23:07	7440-22-4	
Zinc	0.49	mg/L	0.20	1	04/18/19 13:45	04/18/19 23:07	7440-66-6	
6020 MET ICPMS								
Analytical Method: EPA 6020 Preparation Method: EPA 3050B								
Arsenic	22.0	mg/kg	0.86	1	04/14/19 13:41	04/15/19 10:43	7440-38-2	
Cadmium	1.8	mg/kg	0.43	1	04/14/19 13:41	04/15/19 10:43	7440-43-9	
Chromium	111	mg/kg	1.7	1	04/14/19 13:41	04/15/19 10:43	7440-47-3	
Copper	898	mg/kg	4.3	10	04/14/19 13:41	04/15/19 11:06	7440-50-8	
Lead	25.2	mg/kg	4.3	5	04/14/19 13:41	04/15/19 08:43	7439-92-1	N2
Molybdenum	29.6	mg/kg	0.86	1	04/14/19 13:41	04/15/19 10:43	7439-98-7	N2
Nickel	23.3	mg/kg	0.43	1	04/14/19 13:41	04/15/19 10:43	7440-02-0	
Selenium	5.3	mg/kg	0.86	1	04/14/19 13:41	04/15/19 10:43	7782-49-2	
Silver	3.2	mg/kg	0.43	1	04/14/19 13:41	04/15/19 10:43	7440-22-4	
Zinc	1210	mg/kg	43.0	10	04/14/19 13:41	04/15/19 11:06	7440-66-6	
7470 Mercury, TCLP								
Analytical Method: EPA 7470 Preparation Method: EPA 7470								
Leachate Method/Date: EPA 1311; 04/17/19 13:50 Initial pH: 7.91; Final pH: 5.09								
Mercury	<0.0020	mg/L	0.0020	1	04/18/19 09:50	04/18/19 20:56	7439-97-6	
7471 Mercury								
Analytical Method: EPA 7471 Preparation Method: EPA 7471								
Mercury	<1.7	mg/kg	1.7	1	04/15/19 10:52	04/15/19 20:17	7439-97-6	

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50221685

Sample: Digester #4 **Lab ID: 50221685001** Collected: 04/08/19 13:05 Received: 04/09/19 09:48 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270 SVOC SS Soil Analytical Method: EPA 8270 Preparation Method: EPA 3546								
Acenaphthene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	83-32-9	
Acenaphthylene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	208-96-8	
Anthracene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	120-12-7	
Benzo(a)anthracene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	56-55-3	
Benzo(a)pyrene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	50-32-8	
Benzo(b)fluoranthene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	205-99-2	
Benzo(g,h,i)perylene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	191-24-2	
Benzo(k)fluoranthene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	207-08-9	
Benzyl alcohol	<171000	ug/kg	171000	1	04/15/19 12:15	04/15/19 19:36	100-51-6	
4-Bromophenylphenyl ether	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	101-55-3	
Butylbenzylphthalate	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	85-68-7	
4-Chloro-3-methylphenol	<171000	ug/kg	171000	1	04/15/19 12:15	04/15/19 19:36	59-50-7	
4-Chloroaniline	<171000	ug/kg	171000	1	04/15/19 12:15	04/15/19 19:36	106-47-8	
bis(2-Chloroethoxy)methane	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	111-91-1	
bis(2-Chloroethyl) ether	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	111-44-4	
bis(2chloro1methylethyl) ether	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	108-60-1	
2-Chloronaphthalene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	91-58-7	
2-Chlorophenol	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	95-57-8	
4-Chlorophenylphenyl ether	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	7005-72-3	
Chrysene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	218-01-9	
Dibenz(a,h)anthracene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	53-70-3	
Dibenzofuran	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	132-64-9	
3,3'-Dichlorobenzidine	<171000	ug/kg	171000	1	04/15/19 12:15	04/15/19 19:36	91-94-1	
2,4-Dichlorophenol	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	120-83-2	
Diethylphthalate	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	84-66-2	
2,4-Dimethylphenol	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	105-67-9	
Dimethylphthalate	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	131-11-3	
Di-n-butylphthalate	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	84-74-2	
4,6-Dinitro-2-methylphenol	<171000	ug/kg	171000	1	04/15/19 12:15	04/15/19 19:36	534-52-1	
2,4-Dinitrophenol	<416000	ug/kg	416000	1	04/15/19 12:15	04/15/19 19:36	51-28-5	
2,4-Dinitrotoluene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	121-14-2	
2,6-Dinitrotoluene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	606-20-2	
Di-n-octylphthalate	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	117-84-0	
bis(2-Ethylhexyl)phthalate	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	117-81-7	
Fluoranthene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	206-44-0	
Fluorene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	86-73-7	
Hexachloro-1,3-butadiene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	87-68-3	
Hexachlorobenzene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	118-74-1	
Hexachlorocyclopentadiene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	77-47-4	
Hexachloroethane	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	67-72-1	
Indeno(1,2,3-cd)pyrene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	193-39-5	
Isophorone	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	78-59-1	
1-Methylnaphthalene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	90-12-0	
2-Methylnaphthalene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	91-57-6	
2-Methylphenol(o-Cresol)	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	95-48-7	
3&4-Methylphenol(m&p Cresol)	<171000	ug/kg	171000	1	04/15/19 12:15	04/15/19 19:36		

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50221685

Sample: Digester #4 **Lab ID: 50221685001** Collected: 04/08/19 13:05 Received: 04/09/19 09:48 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270 SVOC SS Soil Analytical Method: EPA 8270 Preparation Method: EPA 3546								
Naphthalene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	91-20-3	
2-Nitroaniline	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	88-74-4	
3-Nitroaniline	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	99-09-2	
4-Nitroaniline	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	100-01-6	
Nitrobenzene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	98-95-3	
2-Nitrophenol	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	88-75-5	
4-Nitrophenol	<416000	ug/kg	416000	1	04/15/19 12:15	04/15/19 19:36	100-02-7	
N-Nitroso-di-n-propylamine	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	621-64-7	
N-Nitrosodiphenylamine	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	86-30-6	
Pentachlorophenol	<416000	ug/kg	416000	1	04/15/19 12:15	04/15/19 19:36	87-86-5	
Phenanthrene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	85-01-8	
Phenol	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	108-95-2	
Pyrene	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	129-00-0	
2,4,5-Trichlorophenol	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	95-95-4	
2,4,6-Trichlorophenol	<85700	ug/kg	85700	1	04/15/19 12:15	04/15/19 19:36	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	64	%.	18-98	1	04/15/19 12:15	04/15/19 19:36	4165-60-0	
Phenol-d5 (S)	66	%.	18-108	1	04/15/19 12:15	04/15/19 19:36	4165-62-2	
2-Fluorophenol (S)	65	%.	16-104	1	04/15/19 12:15	04/15/19 19:36	367-12-4	
2,4,6-Tribromophenol (S)	64	%.	10-114	1	04/15/19 12:15	04/15/19 19:36	118-79-6	
2-Fluorobiphenyl (S)	65	%.	21-96	1	04/15/19 12:15	04/15/19 19:36	321-60-8	
p-Terphenyl-d14 (S)	88	%.	29-124	1	04/15/19 12:15	04/15/19 19:36	1718-51-0	
8270 MSSV TCLP Sep Funnel Analytical Method: EPA 8270 Preparation Method: EPA 3510								
Leachate Method/Date: EPA 1311; 04/17/19 13:50 Initial pH: 7.91; Final pH: 5.09								
1,4-Dichlorobenzene	<0.10	mg/L	0.10	1	04/18/19 11:39	04/19/19 04:54	106-46-7	
2,4-Dinitrotoluene	<0.10	mg/L	0.10	1	04/18/19 11:39	04/19/19 04:54	121-14-2	
Hexachloro-1,3-butadiene	<0.10	mg/L	0.10	1	04/18/19 11:39	04/19/19 04:54	87-68-3	
Hexachlorobenzene	<0.10	mg/L	0.10	1	04/18/19 11:39	04/19/19 04:54	118-74-1	
Hexachloroethane	<0.10	mg/L	0.10	1	04/18/19 11:39	04/19/19 04:54	67-72-1	
2-Methylphenol(o-Cresol)	<0.10	mg/L	0.10	1	04/18/19 11:39	04/19/19 04:54	95-48-7	
3&4-Methylphenol(m&p Cresol)	<0.20	mg/L	0.20	1	04/18/19 11:39	04/19/19 04:54		
Nitrobenzene	<0.10	mg/L	0.10	1	04/18/19 11:39	04/19/19 04:54	98-95-3	
Pentachlorophenol	<0.50	mg/L	0.50	1	04/18/19 11:39	04/19/19 04:54	87-86-5	
Pyridine	<0.10	mg/L	0.10	1	04/18/19 11:39	04/19/19 04:54	110-86-1	L2
2,4,5-Trichlorophenol	<0.50	mg/L	0.50	1	04/18/19 11:39	04/19/19 04:54	95-95-4	
2,4,6-Trichlorophenol	<0.10	mg/L	0.10	1	04/18/19 11:39	04/19/19 04:54	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	66	%.	27-95	1	04/18/19 11:39	04/19/19 04:54	4165-60-0	
2-Fluorobiphenyl (S)	65	%.	19-93	1	04/18/19 11:39	04/19/19 04:54	321-60-8	
p-Terphenyl-d14 (S)	86	%.	11-147	1	04/18/19 11:39	04/19/19 04:54	1718-51-0	
Phenol-d5 (S)	19	%.	10-42	1	04/18/19 11:39	04/19/19 04:54	4165-62-2	
2-Fluorophenol (S)	33	%.	10-59	1	04/18/19 11:39	04/19/19 04:54	367-12-4	
2,4,6-Tribromophenol (S)	91	%.	33-108	1	04/18/19 11:39	04/19/19 04:54	118-79-6	

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50221685

Sample: Digester #4 **Lab ID: 50221685001** Collected: 04/08/19 13:05 Received: 04/09/19 09:48 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8260 MSV 5030 Low Level		Analytical Method: EPA 8260						
Acetone	<216	mg/kg	216	50		04/22/19 14:30	67-64-1	
Acrolein	<216	mg/kg	216	50		04/22/19 14:30	107-02-8	
Acrylonitrile	<216	mg/kg	216	50		04/22/19 14:30	107-13-1	
Benzene	<10.8	mg/kg	10.8	50		04/22/19 14:30	71-43-2	
Bromodichloromethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	75-27-4	
Bromoform	<10.8	mg/kg	10.8	50		04/22/19 14:30	75-25-2	
Bromomethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	74-83-9	
2-Butanone (MEK)	<54.1	mg/kg	54.1	50		04/22/19 14:30	78-93-3	
Carbon disulfide	<21.6	mg/kg	21.6	50		04/22/19 14:30	75-15-0	
Carbon tetrachloride	<10.8	mg/kg	10.8	50		04/22/19 14:30	56-23-5	
Chlorobenzene	<10.8	mg/kg	10.8	50		04/22/19 14:30	108-90-7	
Chloroethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	75-00-3	
Chloroform	<10.8	mg/kg	10.8	50		04/22/19 14:30	67-66-3	
Chloromethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	74-87-3	
1,2-Dibromo-3-chloropropane	<21.6	mg/kg	21.6	50		04/22/19 14:30	96-12-8	
Dibromochloromethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	124-48-1	
1,2-Dibromoethane (EDB)	<10.8	mg/kg	10.8	50		04/22/19 14:30	106-93-4	
Dibromomethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	74-95-3	
1,2-Dichlorobenzene	<10.8	mg/kg	10.8	50		04/22/19 14:30	95-50-1	
1,3-Dichlorobenzene	<10.8	mg/kg	10.8	50		04/22/19 14:30	541-73-1	
1,4-Dichlorobenzene	<10.8	mg/kg	10.8	50		04/22/19 14:30	106-46-7	
trans-1,4-Dichloro-2-butene	<216	mg/kg	216	50		04/22/19 14:30	110-57-6	
Dichlorodifluoromethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	75-71-8	
1,1-Dichloroethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	75-34-3	
1,2-Dichloroethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	107-06-2	
1,1-Dichloroethene	<10.8	mg/kg	10.8	50		04/22/19 14:30	75-35-4	
cis-1,2-Dichloroethene	<10.8	mg/kg	10.8	50		04/22/19 14:30	156-59-2	
trans-1,2-Dichloroethene	<10.8	mg/kg	10.8	50		04/22/19 14:30	156-60-5	
1,2-Dichloropropane	<10.8	mg/kg	10.8	50		04/22/19 14:30	78-87-5	
cis-1,3-Dichloropropene	<10.8	mg/kg	10.8	50		04/22/19 14:30	10061-01-5	
trans-1,3-Dichloropropene	<10.8	mg/kg	10.8	50		04/22/19 14:30	10061-02-6	
Ethylbenzene	<10.8	mg/kg	10.8	50		04/22/19 14:30	100-41-4	
Ethyl methacrylate	<216	mg/kg	216	50		04/22/19 14:30	97-63-2	
2-Hexanone	<216	mg/kg	216	50		04/22/19 14:30	591-78-6	
Iodomethane	<216	mg/kg	216	50		04/22/19 14:30	74-88-4	
Methylene Chloride	<43.3	mg/kg	43.3	50		04/22/19 14:30	75-09-2	
Methyl methacrylate	<21.6	mg/kg	21.6	50		04/22/19 14:30	80-62-6	
4-Methyl-2-pentanone (MIBK)	<54.1	mg/kg	54.1	50		04/22/19 14:30	108-10-1	
Methyl-tert-butyl ether	<10.8	mg/kg	10.8	50		04/22/19 14:30	1634-04-4	
Styrene	<10.8	mg/kg	10.8	50		04/22/19 14:30	100-42-5	
1,1,1,2-Tetrachloroethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	630-20-6	
1,1,2,2-Tetrachloroethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	79-34-5	
Tetrachloroethene	<10.8	mg/kg	10.8	50		04/22/19 14:30	127-18-4	
Toluene	<10.8	mg/kg	10.8	50		04/22/19 14:30	108-88-3	
1,1,1-Trichloroethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	71-55-6	
1,1,2-Trichloroethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	79-00-5	

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50221685

Sample: Digester #4 **Lab ID: 50221685001** Collected: 04/08/19 13:05 Received: 04/09/19 09:48 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8260 MSV 5030 Low Level		Analytical Method: EPA 8260						
Trichloroethene	<10.8	mg/kg	10.8	50		04/22/19 14:30	79-01-6	
Trichlorofluoromethane	<10.8	mg/kg	10.8	50		04/22/19 14:30	75-69-4	
1,2,3-Trichloropropane	<10.8	mg/kg	10.8	50		04/22/19 14:30	96-18-4	
Vinyl acetate	<216	mg/kg	216	50		04/22/19 14:30	108-05-4	
Vinyl chloride	<10.8	mg/kg	10.8	50		04/22/19 14:30	75-01-4	
Xylene (Total)	<21.6	mg/kg	21.6	50		04/22/19 14:30	1330-20-7	
Surrogates								
Dibromofluoromethane (S)	102	%.	77-131	50		04/22/19 14:30	1868-53-7	D3
Toluene-d8 (S)	99	%.	77-127	50		04/22/19 14:30	2037-26-5	
4-Bromofluorobenzene (S)	98	%.	65-119	50		04/22/19 14:30	460-00-4	
8260 MSV TCLP		Analytical Method: EPA 5030/8260 Leachate Method/Date: EPA 1311; 04/17/19 13:50						
Benzene	<0.050	mg/L	0.050	1		04/19/19 19:16	71-43-2	
2-Butanone (MEK)	<1.0	mg/L	1.0	1		04/19/19 19:16	78-93-3	
Carbon tetrachloride	<0.050	mg/L	0.050	1		04/19/19 19:16	56-23-5	
Chlorobenzene	<0.050	mg/L	0.050	1		04/19/19 19:16	108-90-7	
Chloroform	<0.050	mg/L	0.050	1		04/19/19 19:16	67-66-3	
1,2-Dichloroethane	<0.050	mg/L	0.050	1		04/19/19 19:16	107-06-2	
1,1-Dichloroethene	<0.050	mg/L	0.050	1		04/19/19 19:16	75-35-4	
Tetrachloroethene	<0.050	mg/L	0.050	1		04/19/19 19:16	127-18-4	
Trichloroethene	<0.050	mg/L	0.050	1		04/19/19 19:16	79-01-6	
Vinyl chloride	<0.020	mg/L	0.020	1		04/19/19 19:16	75-01-4	
Surrogates								
Toluene-d8 (S)	87	%.	85-114	1		04/19/19 19:16	2037-26-5	
4-Bromofluorobenzene (S)	93	%.	85-114	1		04/19/19 19:16	460-00-4	
Dibromofluoromethane (S)	107	%.	80-122	1		04/19/19 19:16	1868-53-7	
Percent Moisture		Analytical Method: SM 2540G						
Percent Moisture	97.7	%	0.10	1		04/12/19 18:10		
2540G Total Percent Solids		Analytical Method: SM 2540G						
Total Solids	2.3	%	0.10	1		04/12/19 18:10		N2
2540G Total Volatile Solids		Analytical Method: SM 2540G						
Total Volatile Solids	58.4	%	0.10	1		04/12/19 18:19		N2
9038 Sulfate Soil		Analytical Method: EPA 9038 Preparation Method: EPA 9038						
Sulfate	<4330	mg/kg	4330	1	04/12/19 14:29	04/17/19 09:01	14808-79-8	N2
9045 pH Soil		Analytical Method: EPA 9045						
pH at 25 Degrees C	8.2	Std. Units	0.10	1		04/11/19 14:08		H3
351.2 Total Kjeldahl Nitrogen		Analytical Method: EPA 351.2 Preparation Method: EPA 351.2						
Nitrogen, Kjeldahl, Total	94400	mg/kg	4270	2	04/12/19 13:20	04/15/19 12:23	7727-37-9	

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50221685

Sample: Digester #4 **Lab ID: 50221685001** Collected: 04/08/19 13:05 Received: 04/09/19 09:48 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
353.2 Nitrogen, NO2/NO3 Analytical Method: EPA 353.2 Preparation Method: EPA 353.2								
Nitrogen, NO2 plus NO3	<215	mg/kg	215	1	04/11/19 10:30	04/11/19 12:17		N2
Nitrogen, Nitrate	<215	mg/kg	215	1	04/11/19 10:30	04/11/19 12:17	14797-55-8	N2
4500 Ammonia Soil, Distilled Analytical Method: SM 4500-NH3 G Preparation Method: SM 4500-NH3 B								
Nitrogen, Ammonia	44500	mg/kg	2140	10	04/18/19 09:45	04/18/19 12:19	7664-41-7	N2
4500 Chloride in Soil Analytical Method: SM 4500-Cl-E Preparation Method: SM 4500-Cl-E								
Chloride	11100	mg/kg	4300	1	04/15/19 12:26	04/19/19 08:37	16887-00-6	N2

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50221685

QC Batch: 495438 Analysis Method: EPA 7470
QC Batch Method: EPA 7470 Analysis Description: 7470 Mercury TCLP
Associated Lab Samples: 50221685001

METHOD BLANK: 2285989 Matrix: Water
Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/L	<0.00067	0.00067	04/18/19 20:52	

LABORATORY CONTROL SAMPLE: 2285990

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	0.005	0.0053	105	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2285991 2285992

Parameter	Units	50221685001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/L	<0.0020	0.015	0.015	0.016	0.016	106	104	75-125	2	20	

MATRIX SPIKE SAMPLE: 2285993

Parameter	Units	5022244001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	<0.0020	4	75-125	M0

MATRIX SPIKE SAMPLE: 2285994

Parameter	Units	50222181001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.016	107	75-125	

MATRIX SPIKE SAMPLE: 2285995

Parameter	Units	50222274001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.017	113	75-125	

MATRIX SPIKE SAMPLE: 2285996

Parameter	Units	50222285001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.016	108	75-125	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

MATRIX SPIKE SAMPLE:		2287159					
		50220518001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Mercury	mg/L	<0.0020	0.015	0.018	118	75-125	

MATRIX SPIKE SAMPLE:		2287160					
		50222348001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Mercury	mg/L	<0.0020	0.015	0.017	114	75-125	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 494656

Analysis Method: EPA 7471

QC Batch Method: EPA 7471

Analysis Description: 7471 Mercury

Associated Lab Samples: 50221685001

METHOD BLANK: 2282809

Matrix: Solid

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/kg	<0.19	0.19	04/15/19 19:56	

LABORATORY CONTROL SAMPLE: 2282810

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	0.52	0.57	111	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2282811 2282812

Parameter	Units	50222046001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	ND	0.55	0.58	0.62	0.65	111	113	75-125	4	20	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 494599

Analysis Method: EPA 6010

QC Batch Method: EPA 3050

Analysis Description: 6010 MET

Associated Lab Samples: 50221685001

METHOD BLANK: 2282522

Matrix: Solid

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Aluminum	mg/kg	<50.0	50.0	04/18/19 12:24	
Calcium	mg/kg	<50.0	50.0	04/18/19 12:24	
Magnesium	mg/kg	<50.0	50.0	04/18/19 12:24	
Phosphorus	mg/kg	ND	50.0	04/18/19 12:24	N2
Potassium	mg/kg	<50.0	50.0	04/18/19 12:24	
Sodium	mg/kg	<50.0	50.0	04/18/19 12:24	

LABORATORY CONTROL SAMPLE: 2282523

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	500	474	95	80-120	
Calcium	mg/kg	500	518	104	80-120	
Magnesium	mg/kg	500	490	98	80-120	
Phosphorus	mg/kg	50	50.4	101	80-120	N2
Potassium	mg/kg	500	479	96	80-120	
Sodium	mg/kg	500	485	97	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2282524 2282525

Parameter	Units	50221872002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	16500	622	616	19600	23200	490	1080	75-125	17	20	E,P6
Calcium	mg/kg	4610	622	616	9990	4750	865	23	75-125	71	20	2d,P6
Magnesium	mg/kg	3460	622	616	4300	4620	135	189	75-125	7	20	P6
Phosphorus	mg/kg	288	62.2	61.6	355	315	108	43	75-125	12	20	N2,P6
Potassium	mg/kg	1250	622	616	2200	2460	152	197	75-125	11	20	M3
Sodium	mg/kg	111	622	616	611	603	80	80	75-125	1	20	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 495496

Analysis Method: EPA 6010

QC Batch Method: EPA 3010

Analysis Description: 6010 MET TCLP

Associated Lab Samples: 50221685001

METHOD BLANK: 2286204

Matrix: Water

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/L	<0.010	0.010	04/18/19 23:02	
Barium	mg/L	<0.50	0.50	04/18/19 23:02	
Cadmium	mg/L	<0.0050	0.0050	04/18/19 23:02	
Chromium	mg/L	<0.010	0.010	04/18/19 23:02	
Lead	mg/L	<0.010	0.010	04/18/19 23:02	
Selenium	mg/L	<0.010	0.010	04/18/19 23:02	
Silver	mg/L	<0.010	0.010	04/18/19 23:02	
Zinc	mg/L	<0.020	0.020	04/18/19 23:02	

LABORATORY CONTROL SAMPLE: 2286205

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	1	0.97	97	80-120	
Barium	mg/L	1	0.99	99	80-120	
Cadmium	mg/L	1	0.96	96	80-120	
Chromium	mg/L	1	0.99	99	80-120	
Lead	mg/L	1	0.91	91	80-120	
Selenium	mg/L	1	0.97	97	80-120	
Silver	mg/L	0.5	0.49	97	80-120	
Zinc	mg/L	1	0.97	97	80-120	

MATRIX SPIKE SAMPLE: 2286206

Parameter	Units	50221685001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	<0.10	10	9.8	98	50-150	
Barium	mg/L	<5.0	10	9.5	94	50-150	
Cadmium	mg/L	<0.050	10	9.5	95	50-150	
Chromium	mg/L	<0.10	10	9.6	96	50-150	
Lead	mg/L	<0.10	10	8.7	87	50-150	
Selenium	mg/L	<0.10	10	9.8	98	50-150	
Silver	mg/L	<0.10	5	4.7	93	50-150	
Zinc	mg/L	0.49	10	10.1	96	50-150	

MATRIX SPIKE SAMPLE: 2286207

Parameter	Units	5022244001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	3.2	10	13.1	99	50-150	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50221685

MATRIX SPIKE SAMPLE:		2286207					
Parameter	Units	5022244001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Barium	mg/L	ND	10	9.6	96	50-150	
Cadmium	mg/L	ND	10	9.6	96	50-150	
Chromium	mg/L	ND	10	9.8	98	50-150	
Lead	mg/L	ND	10	9.1	91	50-150	
Selenium	mg/L	ND	10	9.7	97	50-150	
Silver	mg/L	ND	5	4.8	96	50-150	
Zinc	mg/L	ND	10	9.6	96	50-150	

MATRIX SPIKE SAMPLE:		2286208					
Parameter	Units	50222181001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.9	99	50-150	
Barium	mg/L	ND	10	10.7	96	50-150	
Cadmium	mg/L	ND	10	9.7	97	50-150	
Chromium	mg/L	ND	10	9.8	98	50-150	
Lead	mg/L	ND	10	8.8	88	50-150	
Selenium	mg/L	ND	10	10	100	50-150	
Silver	mg/L	ND	5	4.8	96	50-150	
Zinc	mg/L	3820 ug/L	10	13.5	97	50-150	

MATRIX SPIKE SAMPLE:		2286209					
Parameter	Units	50222274001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.0	100	50-150	
Barium	mg/L	ND	10	10	96	50-150	
Cadmium	mg/L	ND	10	9.8	98	50-150	
Chromium	mg/L	ND	10	9.9	99	50-150	
Lead	mg/L	ND	10	8.8	88	50-150	
Selenium	mg/L	ND	10	10	100	50-150	
Silver	mg/L	ND	5	4.9	98	50-150	
Zinc	mg/L	1490 ug/L	10	11.2	97	50-150	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE:		2286210			2286211							
Parameter	Units	50222256001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/L	ND	10	10	10.1	10.2	101	102	50-150	1	20	
Barium	mg/L	ND	10	10	10.9	11.0	98	98	50-150	0	20	
Cadmium	mg/L	ND	10	10	9.9	10.0	99	100	50-150	1	20	
Chromium	mg/L	ND	10	10	10.1	10.1	101	101	50-150	0	20	
Lead	mg/L	ND	10	10	9.0	9.2	90	92	50-150	2	20	
Selenium	mg/L	ND	10	10	10.2	10.3	102	103	50-150	2	20	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2286210 2286211												
Parameter	Units	50222256001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Silver	mg/L	ND	5	5	4.9	4.9	98	97	50-150	1	20	
Zinc	mg/L	65400 ug/L	10	10	76.3	74.9	108	95	50-150	2	20	

MATRIX SPIKE SAMPLE: 2286212							
Parameter	Units	50222285001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.8	98	50-150	
Barium	mg/L	ND	10	9.4	94	50-150	
Cadmium	mg/L	ND	10	9.8	98	50-150	
Chromium	mg/L	ND	10	10.3	103	50-150	
Lead	mg/L	ND	10	9.5	95	50-150	
Selenium	mg/L	ND	10	9.9	99	50-150	
Silver	mg/L	ND	5	4.7	94	50-150	
Zinc	mg/L	2910 ug/L	10	12.7	98	50-150	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50221685

QC Batch: 494451	Analysis Method: EPA 6020
QC Batch Method: EPA 3050B	Analysis Description: 6020 MET
Associated Lab Samples: 50221685001	

METHOD BLANK: 2281889 Matrix: Solid
Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	<0.10	0.10	04/15/19 08:25	
Cadmium	mg/kg	<0.050	0.050	04/15/19 08:25	
Chromium	mg/kg	<0.20	0.20	04/15/19 08:25	
Copper	mg/kg	<0.050	0.050	04/15/19 08:25	
Lead	mg/kg	<0.10	0.10	04/15/19 08:25	N2
Molybdenum	mg/kg	<0.10	0.10	04/15/19 08:25	N2
Nickel	mg/kg	<0.050	0.050	04/15/19 08:25	
Selenium	mg/kg	<0.10	0.10	04/15/19 08:25	
Silver	mg/kg	<0.050	0.050	04/15/19 08:25	
Zinc	mg/kg	<0.50	0.50	04/15/19 08:25	

LABORATORY CONTROL SAMPLE: 2281890

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	4	3.7	93	80-120	
Cadmium	mg/kg	4	3.9	98	80-120	
Chromium	mg/kg	4	4.2	104	80-120	
Copper	mg/kg	4	4.1	102	80-120	
Lead	mg/kg	4	4.1	104	80-120	N2
Molybdenum	mg/kg	4	4.1	103	80-120	N2
Nickel	mg/kg	4	4.0	101	80-120	
Selenium	mg/kg	4	3.9	98	80-120	
Silver	mg/kg	4	4.1	102	80-120	
Zinc	mg/kg	4	3.9	96	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2281891 2281892

Parameter	Units	50221760001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	<0.097	4	3.9	3.2	3.1	79	78	75-125	3	20	
Cadmium	mg/kg	<0.048	4	3.9	3.9	3.8	96	97	75-125	1	20	
Chromium	mg/kg	0.40	4	3.9	4.4	4.3	101	100	75-125	3	20	
Copper	mg/kg	0.31	4	3.9	4.2	4.1	98	97	75-125	2	20	
Lead	mg/kg	0.18	4	3.9	4.4	4.3	105	105	75-125	3	20	N2
Molybdenum	mg/kg	<0.097	4	3.9	3.5	3.4	87	85	75-125	4	20	N2
Nickel	mg/kg	0.14	4	3.9	4.0	3.9	98	96	75-125	4	20	
Selenium	mg/kg	0.13	4	3.9	3.5	3.4	85	85	75-125	2	20	
Silver	mg/kg	<0.048	4	3.9	3.9	3.9	99	99	75-125	2	20	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2281891 2281892												
Parameter	Units	50221760001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	Max RPD	Max RPD	Qual
Zinc	mg/kg	1.8	4	3.9	5.8	5.5	100	93	75-125	6	20	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 496129

Analysis Method: EPA 8260

QC Batch Method: EPA 8260

Analysis Description: 8260 MSV 5030 Low

Associated Lab Samples: 50221685001

METHOD BLANK: 2289097

Matrix: Solid

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
1,1,1-Trichloroethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
1,1,2,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
1,1,2-Trichloroethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
1,1-Dichloroethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
1,1-Dichloroethene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
1,2,3-Trichloropropane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
1,2-Dibromo-3-chloropropane	mg/kg	<0.010	0.010	04/22/19 13:55	
1,2-Dibromoethane (EDB)	mg/kg	<0.0050	0.0050	04/22/19 13:55	
1,2-Dichlorobenzene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
1,2-Dichloroethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
1,2-Dichloropropane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
1,3-Dichlorobenzene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
1,4-Dichlorobenzene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
2-Butanone (MEK)	mg/kg	<0.025	0.025	04/22/19 13:55	
2-Hexanone	mg/kg	<0.10	0.10	04/22/19 13:55	
4-Methyl-2-pentanone (MIBK)	mg/kg	<0.025	0.025	04/22/19 13:55	
Acetone	mg/kg	<0.10	0.10	04/22/19 13:55	
Acrolein	mg/kg	<0.10	0.10	04/22/19 13:55	
Acrylonitrile	mg/kg	<0.10	0.10	04/22/19 13:55	
Benzene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Bromodichloromethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Bromoform	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Bromomethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Carbon disulfide	mg/kg	<0.010	0.010	04/22/19 13:55	
Carbon tetrachloride	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Chlorobenzene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Chloroethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Chloroform	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Chloromethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
cis-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
cis-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Dibromochloromethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Dibromomethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Dichlorodifluoromethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Ethyl methacrylate	mg/kg	<0.10	0.10	04/22/19 13:55	
Ethylbenzene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Iodomethane	mg/kg	<0.10	0.10	04/22/19 13:55	
Methyl methacrylate	mg/kg	<0.010	0.010	04/22/19 13:55	
Methyl-tert-butyl ether	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Methylene Chloride	mg/kg	<0.020	0.020	04/22/19 13:55	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50221685

METHOD BLANK: 2289097

Matrix: Solid

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Styrene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Tetrachloroethene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Toluene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
trans-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
trans-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
trans-1,4-Dichloro-2-butene	mg/kg	<0.10	0.10	04/22/19 13:55	
Trichloroethene	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Trichlorofluoromethane	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Vinyl acetate	mg/kg	<0.10	0.10	04/22/19 13:55	
Vinyl chloride	mg/kg	<0.0050	0.0050	04/22/19 13:55	
Xylene (Total)	mg/kg	<0.010	0.010	04/22/19 13:55	
4-Bromofluorobenzene (S)	%	98	65-119	04/22/19 13:55	
Dibromofluoromethane (S)	%	100	77-131	04/22/19 13:55	
Toluene-d8 (S)	%	100	77-127	04/22/19 13:55	

LABORATORY CONTROL SAMPLE: 2289098

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	0.1	0.084	84	81-122	
1,1,1-Trichloroethane	mg/kg	0.1	0.086	86	72-125	
1,1,2,2-Tetrachloroethane	mg/kg	0.1	0.090	90	70-124	
1,1,2-Trichloroethane	mg/kg	0.1	0.090	90	77-122	
1,1-Dichloroethane	mg/kg	0.1	0.084	84	69-116	
1,1-Dichloroethene	mg/kg	0.1	0.082	82	70-127	
1,2,3-Trichloropropane	mg/kg	0.1	0.090	90	71-124	
1,2-Dibromo-3-chloropropane	mg/kg	0.1	0.10	103	69-132	
1,2-Dibromoethane (EDB)	mg/kg	0.1	0.090	90	77-126	
1,2-Dichlorobenzene	mg/kg	0.1	0.086	86	73-115	
1,2-Dichloroethane	mg/kg	0.1	0.081	81	72-120	
1,2-Dichloropropane	mg/kg	0.1	0.093	93	77-125	
1,3-Dichlorobenzene	mg/kg	0.1	0.086	86	66-115	
1,4-Dichlorobenzene	mg/kg	0.1	0.082	82	66-114	
2-Butanone (MEK)	mg/kg	0.5	0.45	91	57-145	
2-Hexanone	mg/kg	0.5	0.59	118	64-127	
4-Methyl-2-pentanone (MIBK)	mg/kg	0.5	0.47	95	60-123	
Acetone	mg/kg	0.5	0.44	88	33-174	
Acrolein	mg/kg	2	1.9	94	11-200	
Acrylonitrile	mg/kg	0.4	0.34	85	64-123	
Benzene	mg/kg	0.1	0.081	81	74-119	
Bromodichloromethane	mg/kg	0.1	0.086	86	73-120	
Bromoform	mg/kg	0.1	0.086	86	65-118	
Bromomethane	mg/kg	0.1	0.14	137	37-160	
Carbon disulfide	mg/kg	0.1	0.074	74	65-123	
Carbon tetrachloride	mg/kg	0.1	0.083	83	71-125	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

LABORATORY CONTROL SAMPLE: 2289098

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chlorobenzene	mg/kg	0.1	0.082	82	76-113	
Chloroethane	mg/kg	0.1	0.070	70	59-148	
Chloroform	mg/kg	0.1	0.083	83	71-117	
Chloromethane	mg/kg	0.1	0.062	62	49-112	
cis-1,2-Dichloroethene	mg/kg	0.1	0.084	84	70-122	
cis-1,3-Dichloropropene	mg/kg	0.1	0.096	96	75-120	
Dibromochloromethane	mg/kg	0.1	0.090	90	78-121	
Dibromomethane	mg/kg	0.1	0.088	88	75-125	
Dichlorodifluoromethane	mg/kg	0.1	0.074	74	34-163	
Ethyl methacrylate	mg/kg	0.4	0.35	88	63-132	
Ethylbenzene	mg/kg	0.1	0.084	84	73-118	
Iodomethane	mg/kg	0.2	0.20	100	71-143	
Methyl methacrylate	mg/kg	0.1	0.089	89	50-150	
Methyl-tert-butyl ether	mg/kg	0.1	0.091	91	74-131	
Methylene Chloride	mg/kg	0.1	0.076	76	67-128	
Styrene	mg/kg	0.1	0.089	89	75-120	
Tetrachloroethene	mg/kg	0.1	0.078	78	70-116	
Toluene	mg/kg	0.1	0.081	81	72-112	
trans-1,2-Dichloroethene	mg/kg	0.1	0.084	84	70-120	
trans-1,3-Dichloropropene	mg/kg	0.1	0.091	91	67-119	
trans-1,4-Dichloro-2-butene	mg/kg	0.4	0.44	111	57-124	
Trichloroethene	mg/kg	0.1	0.083	83	74-120	
Trichlorofluoromethane	mg/kg	0.1	0.071	71	59-139	
Vinyl acetate	mg/kg	0.4	0.29	71	70-134	
Vinyl chloride	mg/kg	0.1	0.078	78	58-133	
Xylene (Total)	mg/kg	0.3	0.25	83	71-119	
4-Bromofluorobenzene (S)	%			103	65-119	
Dibromofluoromethane (S)	%			99	77-131	
Toluene-d8 (S)	%			100	77-127	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 495876

Analysis Method: EPA 5030/8260

QC Batch Method: EPA 5030/8260

Analysis Description: 8260 MSV TCLP

Associated Lab Samples: 50221685001

METHOD BLANK: 2287977

Matrix: Water

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1-Dichloroethene	mg/L	<0.050	0.050	04/19/19 14:29	
1,2-Dichloroethane	mg/L	<0.050	0.050	04/19/19 14:29	
2-Butanone (MEK)	mg/L	<1.0	1.0	04/19/19 14:29	
Benzene	mg/L	<0.050	0.050	04/19/19 14:29	
Carbon tetrachloride	mg/L	<0.050	0.050	04/19/19 14:29	
Chlorobenzene	mg/L	<0.050	0.050	04/19/19 14:29	
Chloroform	mg/L	<0.050	0.050	04/19/19 14:29	
Tetrachloroethene	mg/L	<0.050	0.050	04/19/19 14:29	
Trichloroethene	mg/L	<0.050	0.050	04/19/19 14:29	
Vinyl chloride	mg/L	<0.020	0.020	04/19/19 14:29	
4-Bromofluorobenzene (S)	%	93	85-114	04/19/19 14:29	
Dibromofluoromethane (S)	%	106	80-122	04/19/19 14:29	
Toluene-d8 (S)	%	94	85-114	04/19/19 14:29	

LABORATORY CONTROL SAMPLE: 2287978

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	0.5	0.55	110	71-126	
1,2-Dichloroethane	mg/L	0.5	0.46	92	68-119	
2-Butanone (MEK)	mg/L	2.5	2.4	95	62-140	
Benzene	mg/L	0.5	0.51	102	78-117	
Carbon tetrachloride	mg/L	0.5	0.47	95	68-132	
Chlorobenzene	mg/L	0.5	0.49	98	79-113	
Chloroform	mg/L	0.5	0.49	97	73-118	
Tetrachloroethene	mg/L	0.5	0.52	103	76-124	
Trichloroethene	mg/L	0.5	0.53	106	76-120	
Vinyl chloride	mg/L	0.5	0.56	113	70-136	
4-Bromofluorobenzene (S)	%			92	85-114	
Dibromofluoromethane (S)	%			93	80-122	
Toluene-d8 (S)	%			97	85-114	

MATRIX SPIKE SAMPLE: 2287979

Parameter	Units	50221851001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	ND	5	5.6	112	46-148	
1,2-Dichloroethane	mg/L	ND	5	4.8	95	44-138	
2-Butanone (MEK)	mg/L	ND	25	25.9	103	36-153	
Benzene	mg/L	ND	5	5.2	105	49-140	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50221685

MATRIX SPIKE SAMPLE:		2287979					
Parameter	Units	50221851001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Carbon tetrachloride	mg/L	ND	5	4.7	93	45-148	
Chlorobenzene	mg/L	ND	5	4.9	97	47-135	
Chloroform	mg/L	ND	5	4.9	98	49-136	
Tetrachloroethene	mg/L	ND	5	4.9	98	41-145	
Trichloroethene	mg/L	ND	5	5.3	106	43-147	
Vinyl chloride	mg/L	ND	5	5.7	114	49-153	
4-Bromofluorobenzene (S)	%				96	85-114	
Dibromofluoromethane (S)	%				95	80-122	
Toluene-d8 (S)	%				92	85-114	

MATRIX SPIKE SAMPLE:		2287980					
Parameter	Units	50221685001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	<0.050	0.5	0.54	109	46-148	
1,2-Dichloroethane	mg/L	<0.050	0.5	0.45	91	44-138	
2-Butanone (MEK)	mg/L	<1.0	2.5	2.4	98	36-153	
Benzene	mg/L	<0.050	0.5	0.50	100	49-140	
Carbon tetrachloride	mg/L	<0.050	0.5	0.45	91	45-148	
Chlorobenzene	mg/L	<0.050	0.5	0.48	96	47-135	
Chloroform	mg/L	<0.050	0.5	0.47	95	49-136	
Tetrachloroethene	mg/L	<0.050	0.5	0.49	97	41-145	
Trichloroethene	mg/L	<0.050	0.5	0.50	100	43-147	
Vinyl chloride	mg/L	<0.020	0.5	0.56	112	49-153	
4-Bromofluorobenzene (S)	%				97	85-114	
Dibromofluoromethane (S)	%				96	80-122	
Toluene-d8 (S)	%				92	85-114	

MATRIX SPIKE SAMPLE:		2287981					
Parameter	Units	50222244001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	ND	2500	2660	107	46-148	
1,2-Dichloroethane	mg/L	ND	2500	2280	91	44-138	
2-Butanone (MEK)	mg/L	ND	12500	12900	103	36-153	
Benzene	mg/L	ND	2500	2520	101	49-140	
Carbon tetrachloride	mg/L	ND	2500	2280	91	45-148	
Chlorobenzene	mg/L	ND	2500	2360	94	47-135	
Chloroform	mg/L	ND	2500	2360	95	49-136	
Tetrachloroethene	mg/L	ND	2500	2340	93	41-145	
Trichloroethene	mg/L	ND	2500	2500	100	43-147	
Vinyl chloride	mg/L	ND	2500	2770	111	49-153	
4-Bromofluorobenzene (S)	%				97	85-114	
Dibromofluoromethane (S)	%				93	80-122	
Toluene-d8 (S)	%				96	85-114	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 494421

Analysis Method: EPA 8081

QC Batch Method: EPA 3546

Analysis Description: 8081 GCS Pesticides

Associated Lab Samples: 50221685001

METHOD BLANK: 2281689

Matrix: Solid

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
4,4'-DDD	ug/kg	<5.0	5.0	04/12/19 21:00	
4,4'-DDE	ug/kg	<5.0	5.0	04/12/19 21:00	
4,4'-DDT	ug/kg	<5.0	5.0	04/12/19 21:00	
Aldrin	ug/kg	<2.5	2.5	04/12/19 21:00	
alpha-BHC	ug/kg	<2.5	2.5	04/12/19 21:00	
alpha-Chlordane	ug/kg	<2.5	2.5	04/12/19 21:00	
beta-BHC	ug/kg	<2.5	2.5	04/12/19 21:00	
Chlordane (Technical)	ug/kg	<49.8	49.8	04/12/19 21:00	
delta-BHC	ug/kg	<2.5	2.5	04/12/19 21:00	
Dieldrin	ug/kg	<5.0	5.0	04/12/19 21:00	
Endosulfan I	ug/kg	<2.5	2.5	04/12/19 21:00	
Endosulfan II	ug/kg	<5.0	5.0	04/12/19 21:00	
Endosulfan sulfate	ug/kg	<5.0	5.0	04/12/19 21:00	
Endrin	ug/kg	<5.0	5.0	04/12/19 21:00	
Endrin aldehyde	ug/kg	<5.0	5.0	04/12/19 21:00	
gamma-BHC (Lindane)	ug/kg	<2.5	2.5	04/12/19 21:00	
gamma-Chlordane	ug/kg	<2.5	2.5	04/12/19 21:00	
Heptachlor	ug/kg	<2.5	2.5	04/12/19 21:00	
Heptachlor epoxide	ug/kg	<2.5	2.5	04/12/19 21:00	
Toxaphene	ug/kg	<49.8	49.8	04/12/19 21:00	
Decachlorobiphenyl (S)	%	82	18-136	04/12/19 21:00	

LABORATORY CONTROL SAMPLE: 2281690

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
4,4'-DDD	ug/kg	19.6	24.4	124	42-185	
4,4'-DDE	ug/kg	19.6	23.7	121	40-191	
4,4'-DDT	ug/kg	19.6	27.7	141	30-196	
Aldrin	ug/kg	9.8	10.6	108	42-177	
alpha-BHC	ug/kg	9.8	11.1	113	30-178	
alpha-Chlordane	ug/kg	9.8	11.1	113	41-177	
beta-BHC	ug/kg	9.8	10.8	110	39-176	
delta-BHC	ug/kg	9.8	9.9	101	23-162	
Dieldrin	ug/kg	19.6	22.6	115	40-183	
Endosulfan I	ug/kg	9.8	11.2	114	42-181	
Endosulfan II	ug/kg	19.6	22.9	117	40-187	
Endosulfan sulfate	ug/kg	19.6	22.3	114	38-181	
Endrin	ug/kg	19.6	23.3	119	33-200	
Endrin aldehyde	ug/kg	19.6	25.2	128	39-178	
gamma-BHC (Lindane)	ug/kg	9.8	11.0	112	39-170	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50221685

LABORATORY CONTROL SAMPLE: 2281690

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
gamma-Chlordane	ug/kg	9.8	10.8	110	41-180	
Heptachlor	ug/kg	9.8	11.1	113	38-183	
Heptachlor epoxide	ug/kg	9.8	11.0	113	41-181	
Decachlorobiphenyl (S)	%.			154	18-136	S0

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2281699 2281700

Parameter	Units	50221789002	MS	MSD	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	Max RPD	RPD	Qual
		Result	Spike Conc.	Spike Conc.								
4,4'-DDD	ug/kg	<0.0044 mg/kg	74.1	75.5	92.8	98.4	125	130	10-199	6	20	
4,4'-DDE	ug/kg	<0.0054 mg/kg	74.1	75.5	86.7	93.7	117	124	10-200	8	20	
4,4'-DDT	ug/kg	<0.0058 mg/kg	74.1	75.5	178	188	241	250	10-186	5	20	M0
Aldrin	ug/kg	<0.0024 mg/kg	37	37.7	38.0	43.4	103	115	10-184	13	20	
alpha-BHC	ug/kg	<0.0023 mg/kg	37	37.7	39.7	44.6	107	118	15-165	12	20	
alpha-Chlordane	ug/kg	<0.0035 mg/kg	37	37.7	40.5	44.4	109	118	10-182	9	20	
beta-BHC	ug/kg	<0.0032 mg/kg	37	37.7	40.4	43.8	109	116	10-180	8	20	
delta-BHC	ug/kg	<0.0026 mg/kg	37	37.7	36.4	38.6	98	102	10-176	6	20	
Dieldrin	ug/kg	<0.0046 mg/kg	74.1	75.5	82.3	89.3	111	118	10-186	8	20	
Endosulfan I	ug/kg	<0.0022 mg/kg	37	37.7	40.5	44.1	109	117	10-192	9	20	
Endosulfan II	ug/kg	<0.0055 mg/kg	74.1	75.5	85.0	89.1	115	118	10-190	5	20	
Endosulfan sulfate	ug/kg	<0.0043 mg/kg	74.1	75.5	86.6	87.9	117	116	10-191	1	20	
Endrin	ug/kg	<0.0048 mg/kg	74.1	75.5	85.3	92.4	115	122	10-183	8	20	
Endrin aldehyde	ug/kg	<0.0060 mg/kg	74.1	75.5	97.8	99.6	132	132	10-180	2	20	
gamma-BHC (Lindane)	ug/kg	<0.0022 mg/kg	37	37.7	41.4	47.0	112	125	10-174	13	20	
gamma-Chlordane	ug/kg	<0.0023 mg/kg	37	37.7	46.3	50.4	125	134	10-187	8	20	
Heptachlor	ug/kg	<0.0032 mg/kg	37	37.7	44.6	50.8	120	135	10-197	13	20	
Heptachlor epoxide	ug/kg	<0.0025 mg/kg	37	37.7	45.1	52.2	122	138	10-199	14	20	
Decachlorobiphenyl (S)	%.						75	78	18-136			

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 496009

Analysis Method: EPA 8081

QC Batch Method: EPA 3510

Analysis Description: 8081 GCS TCLP Pesticides

Associated Lab Samples: 50221685001

METHOD BLANK: 2288787

Matrix: Water

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chlordane (Technical)	mg/L	<0.00050	0.00050	04/22/19 15:20	
Endrin	mg/L	<0.00010	0.00010	04/22/19 15:20	
gamma-BHC (Lindane)	mg/L	<0.000050	0.000050	04/22/19 15:20	
Heptachlor	mg/L	<0.000050	0.000050	04/22/19 15:20	
Heptachlor epoxide	mg/L	<0.000050	0.000050	04/22/19 15:20	
Methoxychlor	mg/L	<0.00050	0.00050	04/22/19 15:20	
Toxaphene	mg/L	<0.0010	0.0010	04/22/19 15:20	
Decachlorobiphenyl (S)	%	46	18-118	04/22/19 15:20	

LABORATORY CONTROL SAMPLE: 2288788

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Endrin	mg/L	0.002	0.0025	123	41-189	
gamma-BHC (Lindane)	mg/L	0.001	0.0011	108	41-178	
Heptachlor	mg/L	0.001	0.0011	114	10-150	
Heptachlor epoxide	mg/L	0.001	0.0012	123	40-180	
Methoxychlor	mg/L	0.01	0.015	153	34-193	
Decachlorobiphenyl (S)	%			45	18-118	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 494624

Analysis Method: EPA 8082

QC Batch Method: EPA 3546

Analysis Description: 8082 GCS PCB

Associated Lab Samples: 50221685001

METHOD BLANK: 2282694

Matrix: Solid

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	<99.0	99.0	04/15/19 06:30	
PCB-1221 (Aroclor 1221)	ug/kg	<99.0	99.0	04/15/19 06:30	
PCB-1232 (Aroclor 1232)	ug/kg	<99.0	99.0	04/15/19 06:30	
PCB-1242 (Aroclor 1242)	ug/kg	<99.0	99.0	04/15/19 06:30	
PCB-1248 (Aroclor 1248)	ug/kg	<99.0	99.0	04/15/19 06:30	
PCB-1254 (Aroclor 1254)	ug/kg	<99.0	99.0	04/15/19 06:30	
PCB-1260 (Aroclor 1260)	ug/kg	<99.0	99.0	04/15/19 06:30	
Tetrachloro-m-xylene (S)	%.	84	26-140	04/15/19 06:30	

LABORATORY CONTROL SAMPLE: 2282695

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	166	124	75	59-119	
PCB-1260 (Aroclor 1260)	ug/kg	166	134	81	57-119	
Tetrachloro-m-xylene (S)	%.			73	26-140	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2282696

2282697

Parameter	Units	50222135001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
PCB-1016 (Aroclor 1016)	ug/kg	ND	201	200	<121	161	39	80	10-159		20	
PCB-1260 (Aroclor 1260)	ug/kg	0.0092J mg/kg	201	200	<121	140	37	66	11-131		20	
Tetrachloro-m-xylene (S)	%.						45	52	26-140			

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 495440

Analysis Method: EPA 8151A

QC Batch Method: EPA 8151

Analysis Description: 8151 GCS TCLP Herbicides

Associated Lab Samples: 50221685001

METHOD BLANK: 2285999

Matrix: Water

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0010	0.0010	04/18/19 20:22	
2,4-D	mg/L	<0.0010	0.0010	04/18/19 20:22	
2,4-DCAA (S)	%.	67	10-156	04/18/19 20:22	

LABORATORY CONTROL SAMPLE: 2286000

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	0.005	0.0049	98	27-138	
2,4-D	mg/L	0.005	0.0045	90	30-123	
2,4-DCAA (S)	%.			88	10-156	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50221685

QC Batch: 494728	Analysis Method: EPA 8270
QC Batch Method: EPA 3546	Analysis Description: 8270 Solid MSSV Microwave Short Spike
Associated Lab Samples: 50221685001	

METHOD BLANK: 2282933	Matrix: Solid
Associated Lab Samples: 50221685001	

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1-Methylnaphthalene	ug/kg	<329	329	04/15/19 19:04	
2,4,5-Trichlorophenol	ug/kg	<329	329	04/15/19 19:04	
2,4,6-Trichlorophenol	ug/kg	<329	329	04/15/19 19:04	
2,4-Dichlorophenol	ug/kg	<329	329	04/15/19 19:04	
2,4-Dimethylphenol	ug/kg	<329	329	04/15/19 19:04	
2,4-Dinitrophenol	ug/kg	<1590	1590	04/15/19 19:04	
2,4-Dinitrotoluene	ug/kg	<329	329	04/15/19 19:04	
2,6-Dinitrotoluene	ug/kg	<329	329	04/15/19 19:04	
2-Chloronaphthalene	ug/kg	<329	329	04/15/19 19:04	
2-Chlorophenol	ug/kg	<329	329	04/15/19 19:04	
2-Methylnaphthalene	ug/kg	<329	329	04/15/19 19:04	
2-Methylphenol(o-Cresol)	ug/kg	<329	329	04/15/19 19:04	
2-Nitroaniline	ug/kg	<329	329	04/15/19 19:04	
2-Nitrophenol	ug/kg	<329	329	04/15/19 19:04	
3&4-Methylphenol(m&p Cresol)	ug/kg	<658	658	04/15/19 19:04	
3,3'-Dichlorobenzidine	ug/kg	<658	658	04/15/19 19:04	
3-Nitroaniline	ug/kg	<329	329	04/15/19 19:04	
4,6-Dinitro-2-methylphenol	ug/kg	<658	658	04/15/19 19:04	
4-Bromophenylphenyl ether	ug/kg	<329	329	04/15/19 19:04	
4-Chloro-3-methylphenol	ug/kg	<658	658	04/15/19 19:04	
4-Chloroaniline	ug/kg	<658	658	04/15/19 19:04	
4-Chlorophenylphenyl ether	ug/kg	<329	329	04/15/19 19:04	
4-Nitroaniline	ug/kg	<329	329	04/15/19 19:04	
4-Nitrophenol	ug/kg	<1590	1590	04/15/19 19:04	
Acenaphthene	ug/kg	<329	329	04/15/19 19:04	
Acenaphthylene	ug/kg	<329	329	04/15/19 19:04	
Anthracene	ug/kg	<329	329	04/15/19 19:04	
Benzo(a)anthracene	ug/kg	<329	329	04/15/19 19:04	
Benzo(a)pyrene	ug/kg	<329	329	04/15/19 19:04	
Benzo(b)fluoranthene	ug/kg	<329	329	04/15/19 19:04	
Benzo(g,h,i)perylene	ug/kg	<329	329	04/15/19 19:04	
Benzo(k)fluoranthene	ug/kg	<329	329	04/15/19 19:04	
Benzyl alcohol	ug/kg	<658	658	04/15/19 19:04	
bis(2-Chloroethoxy)methane	ug/kg	<329	329	04/15/19 19:04	
bis(2-Chloroethyl) ether	ug/kg	<329	329	04/15/19 19:04	
bis(2-Ethylhexyl)phthalate	ug/kg	<329	329	04/15/19 19:04	
bis(2chloro1methylethyl) ether	ug/kg	<329	329	04/15/19 19:04	
Butylbenzylphthalate	ug/kg	<329	329	04/15/19 19:04	
Chrysene	ug/kg	<329	329	04/15/19 19:04	
Di-n-butylphthalate	ug/kg	<329	329	04/15/19 19:04	
Di-n-octylphthalate	ug/kg	<329	329	04/15/19 19:04	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50221685

METHOD BLANK: 2282933

Matrix: Solid

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Dibenz(a,h)anthracene	ug/kg	<329	329	04/15/19 19:04	
Dibenzofuran	ug/kg	<329	329	04/15/19 19:04	
Diethylphthalate	ug/kg	<329	329	04/15/19 19:04	
Dimethylphthalate	ug/kg	<329	329	04/15/19 19:04	
Fluoranthene	ug/kg	<329	329	04/15/19 19:04	
Fluorene	ug/kg	<329	329	04/15/19 19:04	
Hexachloro-1,3-butadiene	ug/kg	<329	329	04/15/19 19:04	
Hexachlorobenzene	ug/kg	<329	329	04/15/19 19:04	
Hexachlorocyclopentadiene	ug/kg	<329	329	04/15/19 19:04	
Hexachloroethane	ug/kg	<329	329	04/15/19 19:04	
Indeno(1,2,3-cd)pyrene	ug/kg	<329	329	04/15/19 19:04	
Isophorone	ug/kg	<329	329	04/15/19 19:04	
N-Nitroso-di-n-propylamine	ug/kg	<329	329	04/15/19 19:04	
N-Nitrosodiphenylamine	ug/kg	<329	329	04/15/19 19:04	
Naphthalene	ug/kg	<329	329	04/15/19 19:04	
Nitrobenzene	ug/kg	<329	329	04/15/19 19:04	
Pentachlorophenol	ug/kg	<1590	1590	04/15/19 19:04	
Phenanthrene	ug/kg	<329	329	04/15/19 19:04	
Phenol	ug/kg	<329	329	04/15/19 19:04	
Pyrene	ug/kg	<329	329	04/15/19 19:04	
2,4,6-Tribromophenol (S)	%	78	10-114	04/15/19 19:04	
2-Fluorobiphenyl (S)	%	76	21-96	04/15/19 19:04	
2-Fluorophenol (S)	%	80	16-104	04/15/19 19:04	
Nitrobenzene-d5 (S)	%	74	18-98	04/15/19 19:04	
p-Terphenyl-d14 (S)	%	102	29-124	04/15/19 19:04	
Phenol-d5 (S)	%	77	18-108	04/15/19 19:04	

LABORATORY CONTROL SAMPLE: 2282934

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4-Dinitrotoluene	ug/kg	3320	2750	83	32-115	
2-Chlorophenol	ug/kg	3320	2690	81	47-102	
2-Methylnaphthalene	ug/kg	3320	2720	82	49-101	
4-Chloro-3-methylphenol	ug/kg	3320	2840	86	46-112	
4-Nitrophenol	ug/kg	3320	2710	82	28-123	
Acenaphthene	ug/kg	3320	2720	82	45-103	
Acenaphthylene	ug/kg	3320	2820	85	46-103	
Anthracene	ug/kg	3320	2740	82	53-106	
Benzo(a)anthracene	ug/kg	3320	2970	89	52-107	
Benzo(a)pyrene	ug/kg	3320	2870	87	46-108	
Benzo(b)fluoranthene	ug/kg	3320	2830	85	48-111	
Benzo(g,h,i)perylene	ug/kg	3320	2940	88	44-111	
Benzo(k)fluoranthene	ug/kg	3320	2820	85	49-111	
Chrysene	ug/kg	3320	3160	95	53-109	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50221685

LABORATORY CONTROL SAMPLE: 2282934

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Dibenz(a,h)anthracene	ug/kg	3320	2950	89	47-110	
Fluoranthene	ug/kg	3320	2920	88	50-113	
Fluorene	ug/kg	3320	2880	87	46-107	
Indeno(1,2,3-cd)pyrene	ug/kg	3320	2910	88	46-111	
N-Nitroso-di-n-propylamine	ug/kg	3320	2560	77	40-103	
Naphthalene	ug/kg	3320	2670	80	47-97	
Pentachlorophenol	ug/kg	3320	2170	65	27-124	
Phenanthrene	ug/kg	3320	2860	86	53-107	
Phenol	ug/kg	3320	2690	81	40-104	
Pyrene	ug/kg	3320	2890	87	45-119	
2,4,6-Tribromophenol (S)	%			85	10-114	
2-Fluorobiphenyl (S)	%			82	21-96	
2-Fluorophenol (S)	%			81	16-104	
Nitrobenzene-d5 (S)	%			77	18-98	
p-Terphenyl-d14 (S)	%			106	29-124	
Phenol-d5 (S)	%			80	18-108	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2282935 2282936

Parameter	Units	50221685001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
2,4-Dinitrotoluene	ug/kg	<85700	849000	831000	606000	598000	71	72	10-125	1	20	
2-Chlorophenol	ug/kg	<85700	849000	831000	586000	597000	69	72	14-115	2	20	
2-Methylnaphthalene	ug/kg	<85700	849000	831000	615000	603000	72	72	19-118	2	20	
4-Chloro-3-methylphenol	ug/kg	<171000	849000	831000	678000	674000	80	81	14-127	1	20	
4-Nitrophenol	ug/kg	<416000	849000	831000	614000	613000	72	74	10-137	0	20	
Acenaphthene	ug/kg	<85700	849000	831000	613000	619000	72	74	18-113	1	20	
Acenaphthylene	ug/kg	<85700	849000	831000	626000	625000	74	75	17-113	0	20	
Anthracene	ug/kg	<85700	849000	831000	622000	623000	73	75	22-114	0	20	
Benzo(a)anthracene	ug/kg	<85700	849000	831000	700000	695000	82	84	18-115	1	20	
Benzo(a)pyrene	ug/kg	<85700	849000	831000	648000	660000	76	79	14-109	2	20	
Benzo(b)fluoranthene	ug/kg	<85700	849000	831000	627000	646000	74	78	10-120	3	20	
Benzo(g,h,i)perylene	ug/kg	<85700	849000	831000	683000	708000	80	85	13-109	4	20	
Benzo(k)fluoranthene	ug/kg	<85700	849000	831000	650000	680000	77	82	14-114	4	20	
Chrysene	ug/kg	<85700	849000	831000	724000	742000	85	89	18-119	2	20	
Dibenz(a,h)anthracene	ug/kg	<85700	849000	831000	670000	695000	79	83	12-114	4	20	
Fluoranthene	ug/kg	<85700	849000	831000	649000	672000	77	81	20-120	3	20	
Fluorene	ug/kg	<85700	849000	831000	644000	656000	76	79	17-120	2	20	
Indeno(1,2,3-cd)pyrene	ug/kg	<85700	849000	831000	663000	683000	78	82	13-109	3	20	
N-Nitroso-di-n-propylamine	ug/kg	<85700	849000	831000	554000	560000	65	67	24-108	1	20	
Naphthalene	ug/kg	<85700	849000	831000	586000	586000	69	70	20-112	0	20	
Pentachlorophenol	ug/kg	<416000	849000	831000	<407000	<400000	33	35	10-139		20	
Phenanthrene	ug/kg	<85700	849000	831000	629000	638000	74	77	28-113	1	20	
Phenol	ug/kg	<85700	849000	831000	618000	617000	73	74	10-117	0	20	
Pyrene	ug/kg	<85700	849000	831000	722000	700000	85	84	18-130	3	20	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2282935 2282936												
Parameter	Units	50221685001 Result	MS	MSD	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	Max		
			Spike Conc.	Spike Conc.						RPD	RPD	Qual
2,4,6-Tribromophenol (S)	%.						64	62	10-114			
2-Fluorobiphenyl (S)	%.						71	71	21-96			
2-Fluorophenol (S)	%.						70	71	16-104			
Nitrobenzene-d5 (S)	%.						67	68	18-98			
p-Terphenyl-d14 (S)	%.						95	98	29-124			
Phenol-d5 (S)	%.						70	71	18-108			

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 495441

Analysis Method: EPA 8270

QC Batch Method: EPA 3510

Analysis Description: 8270 TCLP MSSV

Associated Lab Samples: 50221685001

METHOD BLANK: 2286001

Matrix: Water

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,4-Dichlorobenzene	mg/L	<0.010	0.010	04/19/19 03:34	
2,4,5-Trichlorophenol	mg/L	<0.050	0.050	04/19/19 03:34	
2,4,6-Trichlorophenol	mg/L	<0.010	0.010	04/19/19 03:34	
2,4-Dinitrotoluene	mg/L	<0.010	0.010	04/19/19 03:34	
2-Methylphenol(o-Cresol)	mg/L	<0.010	0.010	04/19/19 03:34	
3&4-Methylphenol(m&p Cresol)	mg/L	<0.020	0.020	04/19/19 03:34	
Hexachloro-1,3-butadiene	mg/L	<0.010	0.010	04/19/19 03:34	
Hexachlorobenzene	mg/L	<0.010	0.010	04/19/19 03:34	
Hexachloroethane	mg/L	<0.010	0.010	04/19/19 03:34	
Nitrobenzene	mg/L	<0.010	0.010	04/19/19 03:34	
Pentachlorophenol	mg/L	<0.050	0.050	04/19/19 03:34	
Pyridine	mg/L	<0.010	0.010	04/19/19 03:34	
2,4,6-Tribromophenol (S)	%	90	33-108	04/19/19 03:34	
2-Fluorobiphenyl (S)	%	67	19-93	04/19/19 03:34	
2-Fluorophenol (S)	%	42	10-59	04/19/19 03:34	
Nitrobenzene-d5 (S)	%	77	27-95	04/19/19 03:34	
p-Terphenyl-d14 (S)	%	93	11-147	04/19/19 03:34	
Phenol-d5 (S)	%	23	10-42	04/19/19 03:34	

LABORATORY CONTROL SAMPLE: 2286002

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	0.1	0.057	57	10-83	
2,4,5-Trichlorophenol	mg/L	0.1	0.089	89	39-101	
2,4,6-Trichlorophenol	mg/L	0.1	0.092	92	39-109	
2,4-Dinitrotoluene	mg/L	0.1	0.083	83	39-111	
2-Methylphenol(o-Cresol)	mg/L	0.1	0.058	58	29-86	
3&4-Methylphenol(m&p Cresol)	mg/L	0.2	0.11	55	22-84	
Hexachloro-1,3-butadiene	mg/L	0.1	0.054	54	10-90	
Hexachlorobenzene	mg/L	0.1	0.066	66	31-117	
Hexachloroethane	mg/L	0.1	0.049	49	10-81	
Nitrobenzene	mg/L	0.1	0.078	78	40-96	
Pentachlorophenol	mg/L	0.1	0.057	57	33-121	
Pyridine	mg/L	0.1	<0.010	9	10-52	L2
2,4,6-Tribromophenol (S)	%			96	33-108	
2-Fluorobiphenyl (S)	%			77	19-93	
2-Fluorophenol (S)	%			44	10-59	
Nitrobenzene-d5 (S)	%			77	27-95	
p-Terphenyl-d14 (S)	%			92	11-147	
Phenol-d5 (S)	%			26	10-42	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50221685

MATRIX SPIKE SAMPLE:		2286003					
Parameter	Units	50221851001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	ND	1	<0.50	35	13-72	ED
2,4,5-Trichlorophenol	mg/L	ND	1	<2.5	16	36-101	M1
2,4,6-Trichlorophenol	mg/L	ND	1	<0.50	19	30-106	M1
2,4-Dinitrotoluene	mg/L	ND	1	<0.50	11	36-97	M1
2-Methylphenol(o-Cresol)	mg/L	ND	1	<0.50	23	24-84	M1
3&4-Methylphenol(m&p Cresol)	mg/L	ND	2	<1.0	-21	17-82	M1
Hexachloro-1,3-butadiene	mg/L	ND	1	<0.50	36	10-82	
Hexachlorobenzene	mg/L	ND	1	<0.50	43	20-99	
Hexachloroethane	mg/L	ND	1	<0.50	29	10-73	
Nitrobenzene	mg/L	ND	1	<0.50	41	32-92	
Pentachlorophenol	mg/L	ND	1	<2.5	0	27-122	M1
Pyridine	mg/L	ND	1	<0.50	22	10-55	
2,4,6-Tribromophenol (S)	%.				9	33-108	S0
2-Fluorobiphenyl (S)	%.				44	19-93	
2-Fluorophenol (S)	%.				4	10-59	S0
Nitrobenzene-d5 (S)	%.				39	27-95	
p-Terphenyl-d14 (S)	%.				59	11-147	
Phenol-d5 (S)	%.				2	10-42	S0

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 494105

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: Dry Weight/Percent Moisture

Associated Lab Samples: 50221685001

SAMPLE DUPLICATE: 2279999

Parameter	Units	50221685001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	97.7	97.7	0	5	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 494107

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: 2540G Total Solids

Associated Lab Samples: 50221685001

SAMPLE DUPLICATE: 2280002

Parameter	Units	50221685001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	2.3	2.3	0	10	N2

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 494106

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: 2540G Total Volatile Solids

Associated Lab Samples: 50221685001

SAMPLE DUPLICATE: 2280000

Parameter	Units	50221685001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Volatile Solids	%	58.4	58.6	0	10	N2

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 494477

Analysis Method: EPA 9038

QC Batch Method: EPA 9038

Analysis Description: 9038 Sulfate Soil

Associated Lab Samples: 50221685001

METHOD BLANK: 2282049

Matrix: Solid

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Sulfate	mg/kg	<100	100	04/17/19 09:01	N2

LABORATORY CONTROL SAMPLE: 2282050

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Sulfate	mg/kg	200	189	94	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2282051 2282052

Parameter	Units	50221114001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Sulfate	mg/kg	<763	3830	3830	4270	4180	109	107	90-110	2	20	N2

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 494115

Analysis Method: EPA 9045

QC Batch Method: EPA 9045

Analysis Description: 9045 pH

Associated Lab Samples: 50221685001

SAMPLE DUPLICATE: 2280035

Parameter	Units	50221789001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	8.4	8.4	0	2	H3

SAMPLE DUPLICATE: 2280140

Parameter	Units	50221667001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	7.8	7.7	0	2	H3

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 494164

Analysis Method: EPA 351.2

QC Batch Method: EPA 351.2

Analysis Description: 351.2 TKN

Associated Lab Samples: 50221685001

METHOD BLANK: 2280261

Matrix: Solid

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<49.3	49.3	04/15/19 10:50	

LABORATORY CONTROL SAMPLE: 2280262

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	492	498	101	90-110	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2280263 2280264

Parameter	Units	50221867001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	10500	493	495	9080	13600	-279	627	90-110	40	20	H3,P6, R1

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 494133

Analysis Method: EPA 353.2

QC Batch Method: EPA 353.2

Analysis Description: 353.2 Nitrate + Nitrite

Associated Lab Samples: 50221685001

METHOD BLANK: 2280102

Matrix: Solid

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Nitrate	mg/kg	<5.0	5.0	04/11/19 12:10	N2
Nitrogen, NO2 plus NO3	mg/kg	<5.0	5.0	04/11/19 12:10	N2

LABORATORY CONTROL SAMPLE: 2280103

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Nitrate	mg/kg	10	10.4	104	90-110	N2
Nitrogen, NO2 plus NO3	mg/kg	20	20.4	102	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2280104 2280105

Parameter	Units	50221114001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Nitrate	mg/kg	<38.2	76.7	76.7	82.5	83.7	104	105	90-110	1	20	N2
Nitrogen, NO2 plus NO3	mg/kg	<38.2	153	153	163	164	102	103	90-110	1	20	N2

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 495449

Analysis Method: SM 4500-NH3 G

QC Batch Method: SM 4500-NH3 B

Analysis Description: 4500 Ammonia, Distilled

Associated Lab Samples: 50221685001

METHOD BLANK: 2286038

Matrix: Solid

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Ammonia	mg/kg	<4.9	4.9	04/18/19 12:06	N2

LABORATORY CONTROL SAMPLE: 2286039

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Ammonia	mg/kg	19.9	20.6	104	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2286040 2286041

Parameter	Units	50222319001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Ammonia	mg/kg	1170	162	162	1230	1390	39	138	90-110	12	20	N2,P6

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50221685

QC Batch: 494738

Analysis Method: SM 4500-Cl-E

QC Batch Method: SM 4500-Cl-E

Analysis Description: 4500 Chloride

Associated Lab Samples: 50221685001

METHOD BLANK: 2282970

Matrix: Solid

Associated Lab Samples: 50221685001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	<99.5	99.5	04/19/19 08:34	N2

LABORATORY CONTROL SAMPLE: 2282971

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	199	213	107	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2282972 2282973

Parameter	Units	50221685001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	11100	8620	8620	20300	20500	107	109	90-110	1	20	N2

MATRIX SPIKE SAMPLE: 2282974

Parameter	Units	50221863019 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	ND	281	409	113	90-110	M0,N2

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALIFIERS

Project: Biosolids
Pace Project No.: 50221685

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
ND - Not Detected at or above adjusted reporting limit.
TNTC - Too Numerous To Count
J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
MDL - Adjusted Method Detection Limit.
PQL - Practical Quantitation Limit.
RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
S - Surrogate
1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
LCS(D) - Laboratory Control Sample (Duplicate)
MS(D) - Matrix Spike (Duplicate)
DUP - Sample Duplicate
RPD - Relative Percent Difference
NC - Not Calculable.
SG - Silica Gel - Clean-Up
U - Indicates the compound was analyzed for, but not detected.
N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
TNI - The NELAC Institute.

BATCH QUALIFIERS

Batch: 495440

[M5] A matrix spike/matrix spike duplicate was not performed for this batch due to insufficient sample volume.

Batch: 496009

[M5] A matrix spike/matrix spike duplicate was not performed for this batch due to insufficient sample volume.

ANALYTE QUALIFIERS

1d A matrix spike/matrix spike duplicate was not performed for this batch due to insufficient sample volume.
2d RPD is outside control limits due to sample non-homogeneity.4-19-19
D3 Sample was diluted due to the presence of high levels of non-target analytes or other matrix interference.
E Analyte concentration exceeded the calibration range. The reported result is estimated.
ED Due to the extract's physical characteristics, the analysis was performed at dilution.
H3 Sample was received or analysis requested beyond the recognized method holding time.
L2 Analyte recovery in the laboratory control sample (LCS) was below QC limits. Results for this analyte in associated samples may be biased low.
M0 Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.
M1 Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery.
M3 Matrix spike recovery was outside laboratory control limits due to matrix interferences.
N2 The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
P1 Routine initial sample volume or weight was not used for extraction, resulting in elevated reporting limits.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALIFIERS

Project: Biosolids

Pace Project No.: 50221685

ANALYTE QUALIFIERS

P6	Matrix spike recovery was outside laboratory control limits due to a parent sample concentration notably higher than the spike level.
R1	RPD value was outside control limits.
S0	Surrogate recovery outside laboratory control limits.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Biosolids

Pace Project No.: 50221685

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50221685001	Digester #4	EPA 3546	494421	EPA 8081	494538
50221685001	Digester #4	EPA 3510	496009	EPA 8081	496095
50221685001	Digester #4	EPA 3546	494624	EPA 8082	494664
50221685001	Digester #4	EPA 8151	495440	EPA 8151A	495672
50221685001	Digester #4	EPA 3050	494599	EPA 6010	495418
50221685001	Digester #4	EPA 3010	495496	EPA 6010	495697
50221685001	Digester #4	EPA 3050B	494451	EPA 6020	494672
50221685001	Digester #4	EPA 7470	495438	EPA 7470	495694
50221685001	Digester #4	EPA 7471	494656	EPA 7471	494850
50221685001	Digester #4	EPA 3546	494728	EPA 8270	494844
50221685001	Digester #4	EPA 3510	495441	EPA 8270	495710
50221685001	Digester #4	EPA 8260	496129		
50221685001	Digester #4	EPA 5030/8260	495876		
50221685001	Digester #4	SM 2540G	494105		
50221685001	Digester #4	SM 2540G	494107		
50221685001	Digester #4	SM 2540G	494106		
50221685001	Digester #4	EPA 9038	494477	EPA 9038	494494
50221685001	Digester #4	EPA 9045	494115		
50221685001	Digester #4	EPA 351.2	494164	EPA 351.2	494712
50221685001	Digester #4	EPA 353.2	494133	EPA 353.2	494195
50221685001	Digester #4	SM 4500-NH3 B	495449	SM 4500-NH3 G	495484
50221685001	Digester #4	SM 4500-CI-E	494738	SM 4500-CI-E	495346

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

Section A		Section B		Section C	
Required Client Information:		Required Project Information:		Invoice Information:	
Company:	City of Cadillac	Report To:	Cindy Tomaszewski	Attention:	CINDY TOMASZEWSKI
Address:	1121 Plett Rd	Copy To:		Company Name:	
	Cadillac, MI 49601			Address:	
Email:	ctomaszewski@cadillac-mi.net	Purchase Order #:		Peace Quote:	
Phone:	231-775-2368	Project Name:	Biosolids	Peace Project Manager:	melanie.booms@pacelabs.com,
Requested Due Date:		Project #:		Peace Profile #:	8311
				Regulatory Agency	
				NDPES / MDEQ	
				State / Location	
				A	

[illegible]

ADDITIONAL COMMENTS	RELINQUISHED BY / AFFILIATION	DATE	TIME	ACCEPTED BY / AFFILIATION	DATE	TIME	SAMPLE CONDITIONS
	City of Castille	4-8-19	1:05pm	Ship via UPS Ground	4-9-19	948	

SAMPLER NAME AND SIGNATURE

PRINT Name of SAMPLER: ALYSHA LACZKO

SIGNATURE of SAMPLER: [Signature]

DATE Signed: 4-8-19

TEMP in C

Received on _____

Custody (Y/N) _____

Sealed (Y/N) _____

Cooler (Y/N) _____

Samples (Y/N) _____

Contact (Y/N) _____

Page 49 of 50

Sample Conditions Upon Receipt Form

WO#: 50221685

Date: 4-9-19	Evaluated by: AWS	Affili: MSB	Due Date: 04/23/19
Client:	Project Manager: MSB	Wk PM: MSB	CLIENT: GR-Cadillac
Profile ID:			
Sample Receiving Non Conformance Form Required: YES ☒ NO	Rush Turn Around Time Requested: YES ☒ NO	Due Date:	
Page 1 Of 1	Lab Notified of Rush or Short Holds: YES ☒ NO		

Lab Sample Receipt Checklist:

Samples Received Via:	FEDEX	UPS	CLIENT	PACE COURIER
Custody Seals Present and Intact:	YES	NO	NA	
USDA Regulated Soils:	YES	NO	NA	
Short Holds Present (< 72 Hours):	YES	NO	NA	
Samples Received in Hold:	YES	NO	NA	
Custody Signatures Present:	YES	NO	NA	
Collector Signature Present:	YES	NO	NA	
Samples Received On Ice:	YES	NO	NA	
Type of Ice: WET BLUE DRY NONE	YES	NO	NA	
Packing Material Used:	YES	NO	NA	
IR Gun #: 202 402 Temp should be 0-6°C	Cooler Temp Upon Receipt: 6.0 °C			
Temp Blank Received:	YES	NO	NA	
Trip Blank Received: Type: HCL MeOH TSP OTHER	YES	NO	NA	
Bottles Intact:	YES	NO	NA	
Correct Bottles:	YES	NO	NA	
Sufficient Volume:	YES	NO	NA	
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation Exceptions are VOA, coliform, TOC, O & G, HEM, DRO	YES	NO	NA	pH Strip Lot Number: 1
VOA Headspace Acceptable (<6mm):	YES	NO	NA	
Comments:	N/A			